CHAPTER 3

MOLECULAR BASIS OF INHERITANCE

CHAPTER ESSENCE AT A GLANCE:
DNA double helix codes genetic information, replicated semiconservatively and expressed via transcription and translation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Experimental Identification of Genetic Material:** Griffith (1928) demonstrated bacterial transformation in Streptococcus pneumoniae from heat-killed S to live R strain. Avery, MacLeod, and McCarty (1944) proved transforming principle is DNA, destroyed solely by DNase. Hershey and Chase (1952) provided unequivocal proof using bacteriophage T2 labeled with radioactive 32P (DNA) and 35S (protein), showing only 32P entered bacteria.

• **Semiconservative Replication and Fork Machinery:** Meselson and Stahl (1958) demonstrated semiconservative replication in E. coli using 15N heavy isotope and CsCl density centrifugation: generation 1 yielded hybrid 14N-15N DNA, generation 2 yielded equal hybrid and light 14N-14N DNA. Replication machinery: DNA Polymerase III synthesizes strictly 5' to 3', producing continuous leading strand and discontinuous Okazaki fragments on lagging strand joined by DNA ligase.

• **Transcription and Eukaryotic Processing:** Transcription uses a DNA-dependent RNA polymerase binding promoter with sigma factor, transcribing antisense template strand 3' to 5'. In eukaryotes, pre-mRNA (hnRNA) undergoes post-transcriptional maturation: 5' Capping (methyl guanosine triphosphate), 3' Polyadenylation (tailing with 200-300 adenylate residues), and Splicing (spliceosome excises non-coding introns and ligates coding exons).

• **Genetic Code Features and the Lac Operon:** Genetic code: Triplet, 64 codons (61 sense, 3 stop: UAA, UAG, UGA), degenerate, unambiguous, universal, and non-overlapping; AUG has dual role (initiator codon and codes for methionine). Jacob and Monod Lac Operon in E. coli: Polycistronic structural genes lac z (beta-galactosidase), lac y (permease), and lac a (transacetylase), regulated by lacI repressor protein; allolactose acts as inducer binding repressor to derepress transcription.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Meselson-Stahl Density Centrifugation} = {}^{15}\text{N-}{}^{15}\text{N} \rightarrow {}^{14}\text{N-}{}^{15}\text{N} \rightarrow 50\%\ {}^{14}\text{N-}{}^{15}\text{N} + 50\%\ {}^{14}\text{N-}{}^{14}\text{N}$$

Semiconservative DNA Replication Density Proof

$$\text{Central Dogma} = \text{DNA} \xrightarrow[\text{Replication}]{\text{DNA Pol}} \text{DNA} \xrightarrow[\text{Transcription}]{\text{RNA Pol}} \text{mRNA} \xrightarrow[\text{Translation}]{\text{Ribosome}} \text{Protein}$$

Francis Crick Directional Genetic Flow Paradigm

$$\text{Lac Operon Polycistron} = \text{lac z } (\beta\text{-galactosidase}) + \text{lac y (Permease)} + \text{lac a (Transacetylase)}$$

Structural Gene Cluster of the Inducible Lac Operon

$$\text{Eukaryotic hnRNA Maturation} = 5'\text{-Capping } (\text{m}^7\text{Gppp}) + \text{Splicing (Introns out)} + 3'\text{-Poly(A) Tailing}$$

Post-Transcriptional Processing of Eukaryotic Pre-mRNA

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** DNA polymerase synthesizes new DNA strictly in the 5' to 3' direction; therefore, the template strand running 3' to 5' produces the continuous leading strand.

• **Alert:** AUG has dual functions: it acts as the initiation codon for translation and codes for the amino acid methionine.

• **Alert:** In the Lac Operon, allolactose (not glucose) acts as the inducer; high glucose levels suppress operon expression via catabolite repression.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State any four salient properties of the genetic code. Explain why the genetic code is described as degenerate and unambiguous. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Salient properties: (1) The codon is triplet, (2) One codon codes for only one amino acid, (3) It is nearly universal, (4) AUG has dual functions. **[1 Mark]**

• Degenerate: Some amino acids are coded by more than one codon (e.g., 61 codons code for 20 amino acids). Unambiguous: One particular codon codes for only one specific amino acid. **[1 Mark]**

2. What are Okazaki fragments? Name the enzyme responsible for joining these discontinuous fragments into a continuous strand. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Okazaki fragments: Short, discontinuously synthesized DNA segments formed on the lagging template strand (running 5' to 3' direction into replication fork) during DNA replication. **[1 Mark]**

• Joining enzyme: DNA ligase catalyzes phosphodiester bond formation between adjacent Okazaki fragments to complete the lagging strand. **[1 Mark]**

3. Describe the Hershey and Chase experiment using radioactive 32P and 35S that provided unequivocal proof that DNA is the genetic material. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Radioactive labeling: Bacteriophages were grown in medium with radioactive sulfur 35S (labeled protein coat) and radioactive phosphorus 32P (labeled DNA). **[1 Mark]**

• Infection and blending: Labeled viruses infected E. coli; viral coats were separated from bacteria using a blender, and separated by centrifugation. **[1 Mark]**

• Results: Bacteria infected with 32P showed radioactivity in the pellet, whereas bacteria infected with 35S showed no radioactivity (remained in supernatant), proving that DNA enters bacteria as the genetic material. **[1 Mark]**

4. Explain the Meselson and Stahl experiment using 15N isotope and CsCl density gradient centrifugation, demonstrating semiconservative DNA replication. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Heavy isotope growth: E. coli was grown in 15NH4Cl medium for several generations, incorporating heavy 15N into DNA (15N-15N). **[1 Mark]**

• Generation 1 (20 min): Transferred to 14NH4Cl normal medium; extracted DNA centrifuged in CsCl density gradient formed a single intermediate hybrid density band (14N-15N). **[1 Mark]**

• Generation 2 (40 min): Replicated again in 14N medium; extracted DNA formed two equal bands: 50% light density (14N-14N) and 50% hybrid density (14N-15N), confirming semiconservative replication. **[1 Mark]**

5. Explain the post-transcriptional modifications that convert eukaryotic heterogeneous nuclear RNA (hnRNA) into functional mature mRNA. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Capping: An unusual nucleotide, methyl guanosine triphosphate (m7Gppp), is added to the 5'-end of hnRNA to protect it from exonucleases. **[1 Mark]**

• Tailing: Polyadenylation adds about 200 to 300 adenylate residues to the 3'-end in a template-independent manner to stabilize the transcript. **[1 Mark]**

• Splicing: Non-coding intervening sequences (introns) are removed by the spliceosome, and functional coding sequences (exons) are joined together in a defined sequence. **[1 Mark]**

6. Explain the operational mechanism of the Lac Operon in E. coli in the presence and absence of lactose (inducer). [Score: 3 Marks]

Answer & Key Valuation Steps:

• In absence of inducer: The regulatory lacI gene constantly transcribes repressor mRNA, synthesizing active repressor protein that binds tightly to the operator region, blocking RNA polymerase from transcribing structural genes. **[1.5 Marks]**

• In presence of inducer (lactose or allolactose): Inducer binds to the repressor protein, causing a conformational change that inactivates it; repressor cannot bind operator, allowing RNA polymerase to transcribe lac z, lac y, and lac a structural genes. **[1.5 Marks]**

7. A synthetic mRNA strand has the sequence: 5'-AUG GCA UUU UAA CCU-3'. Deduce: (a) The sequence of the DNA template strand that transcribed it, (b) The anticodon on the tRNA bringing the first amino acid, (c) The number of amino acids in the translated polypeptide before termination, and (d) Name the initiator amino acid. [Score: 4 Marks]

Answer & Key Valuation Steps:

• (a) DNA template strand sequence: The template strand is complementary and antiparallel (running 3' to 5'): 3'-TAC CGT AAA ATT GGA-5'. **[1 Mark]**

• (b) First tRNA anticodon: The initiator codon is 5'-AUG-3'; the complementary anticodon on initiator methionyl-tRNA is 3'-UAC-5'. **[1 Mark]**

• (c) Translated amino acid count: Codons read: AUG (Methionine), GCA (Alanine), UUU (Phenylalanine), followed by UAA which is a stop codon terminating translation; hence exactly 3 amino acids are incorporated. **[1 Mark]**

• (d) Initiator amino acid: Methionine (Met). **[1 Mark]**